PART I: MINING MONITORING DASHBOARD

Load the data from the dataset

IF the tables cannot be loaded

DISPLAY a data-loading error

STOP

END IF

SET workers = Worker Safety records

SET incidents = Safety Incidents records

SET observations = Safety Observations records

SET equipment = Equipment records

SET alerts = Alerts records

CALCULATE total_workers

AS the number of unique Worker IDs

CALCULATE total_incidents

AS the number of unique Incident IDs

CALCULATE near_misses

AS the number of incidents whose Incident Type is "Near Miss"

CALCULATE high_risk_observations

AS the number of observations whose Risk Level is "High" or "Critical"

CALCULATE ppe_compliance

AS the percentage of workers whose PPE Compliance is "Yes"

CALCULATE total_equipment

AS the number of unique Equipment IDs

CALCULATE available_equipment

AS the number of unique Equipment IDs

whose Maintenance Status is not "Under Maintenance"

CALCULATE equipment_under_maintenance

AS the number of equipment records whose Maintenance Status

is "Under Maintenance"

SELECT equipment_alerts

FROM alerts where Source is "Equipment"

CALCULATE total_equipment_alerts

AS the number of equipment alert records

CALCULATE average_equipment_availability

AS the mean of Availability (%) in equipment records

SELECT critical_open_alerts

FROM equipment_alerts where Severity is "Critical"

AND Status is "Open"

CALCULATE critical_equipment_alerts

AS the number of critical_open_alerts

DISPLAY all calculated dashboard indicators

END PART I

PART J: INTERACTIVE DATA ANAYSIS

BEGIN

RECEIVE the application tables

```
DISPLAY the available analysis questions
GET the question selected by the user

IF the selected question is about incidents by department
  GET a date range
  FILTER incidents to the selected date range
  COUNT unique Incident IDs for each Department
  SORT department counts from highest to lowest

  IF no incident counts are available
    DISPLAY "No incidents in the selected period"
  ELSE
    DISPLAY the department with the highest count
    DISPLAY a bar chart and the department-count table
  END IF

ELSE IF the question is about highest downtime or lowest availability
  GET the selected measure: Downtime or Availability
  SORT equipment by that measure

  IF no equipment records are available
    DISPLAY an information message
  ELSE
    DISPLAY the equipment with the highest downtime
    OR the equipment with the lowest availability
    DISPLAY a chart and a table of up to ten equipment records
  END IF

ELSE IF the question is about highest vibration or temperature
  GET the selected measurement
```

```
  CONVERT its values to numeric form
  REMOVE records without a usable value
  SORT equipment from highest reading to lowest

  IF no readings are available
    DISPLAY an information message
  ELSE
    DISPLAY the equipment with the highest reading
    DISPLAY a chart and a table of the ranked equipment
  END IF

ELSE IF the question is about equipment alerts
  SELECT alert records whose Source is "Equipment"
  COUNT unique Alert IDs for each equipment Record ID
  SORT alert counts from highest to lowest

  IF no equipment alerts are available
    DISPLAY an information message
  ELSE
    DISPLAY the equipment with the highest alert count
    DISPLAY a chart and an alert-count table
  END IF

ELSE IF the question is about maintenance priority
  CALCULATE maintenance indicators for each equipment record
  IDENTIFY records with maintenance flags
  SORT flagged equipment by the priority indicators
  DISPLAY the flagged equipment and its maintenance details

  IF no equipment is flagged
    DISPLAY a message that no equipment matches the conditions
```

END IF

ELSE IF the question is about common incident causes
COUNT incident records for each Cause
DISPLAY a chart and a cause-count table

ELSE IF the question is about high-risk observations by department
SELECT observations with Risk Level "High" or "Critical"
COUNT unique Observation IDs for each Department
SORT counts from highest to lowest
DISPLAY a chart and a department-count table

ELSE IF the question is about departments with safety
and equipment concerns
GET a date range for incidents
FILTER incidents to that date range

GET a date range for safety observations
FILTER observations to that date range

BY Department, CALCULATE:
Number of safety incidents
Number of high or critical incidents
Number of high or critical observations
Average equipment downtime
Number of recorded alerts

COMBINE the departmental results
REPLACE missing department values with zero
SORT the comparison by high or critical incidents,
high or critical observations, and average downtime

```
        DISPLAY the departmental comparison table

    ELSE
        ASSESS temperature and vibration condition for each equipment item
        COUNT equipment items for which both readings are classified Normal
        CALCULATE the percentage of such equipment items
        DISPLAY the count, percentage, and condition results
    END IF

END PART J
```